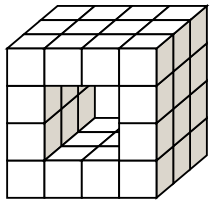
Preview: y2002_q20 (v1)

Counting cubes in a 3-D building, hidden ones included (slice x number of slices)

Part A: Each question is worth 3 points

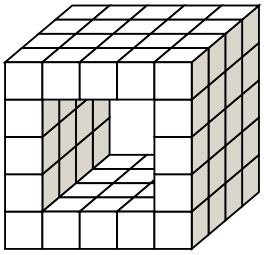
1. Robert built this tunnel out of identical cubes. The hole goes straight through from the front to the back.



How many cubes did Robert use?

A. 12 ***B.*** 24 ***C.*** 26 ***D.*** 36 ***E.*** 48

2. Sara built this tunnel out of identical cubes. The hole goes straight through from the front to the back.



How many cubes did Sara use?

A. 16 ***B.*** 48 ***C.*** 64 ***D.*** 80 ***E.*** 100

3. Nina made a tunnel out of identical cubes (fig. 1). Then Nina took it apart and built a complete pyramid (fig. 2) with some of the cubes. Nothing is hollow inside.

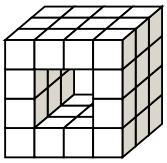


fig. 1

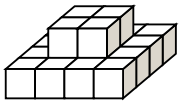


fig. 2

How many cubes from the tunnel were *not* used for the pyramid?

A. 28 ***B.*** 16 ***C.*** 15 ***D.*** 14 ***E.*** 4

4. Leo made a tunnel out of identical cubes (fig. 1). Then Leo took it apart and built a complete pyramid (fig. 2) with some of the cubes. Nothing is hollow inside.

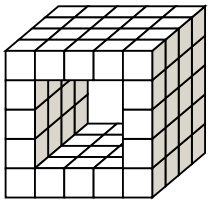


fig. 1

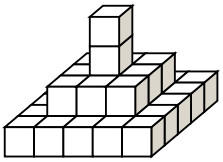


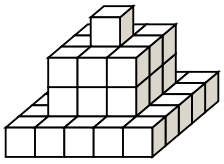
fig. 2

How many cubes from the tunnel were *not* used for the pyramid?

A. 48 ***B.*** 44 ***C.*** 34 ***D.*** 29 ***E.*** 28

Part B: Each question is worth 4 points

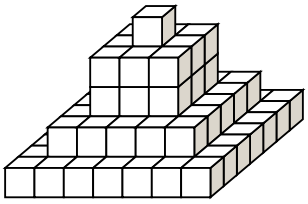
5. Leo built this pyramid out of identical cubes. Every layer is a full square, with no holes hidden inside.



How many cubes are in the pyramid?

A. 12 ***B.*** 24 ***C.*** 35 ***D.*** 43 ***E.*** 44

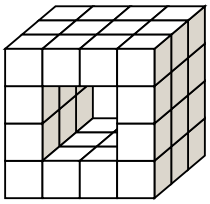
6. Leo built this pyramid out of identical cubes. Every layer is a full square, with no holes hidden inside.



How many cubes are in the pyramid?

A. 110 ***B.*** 106 ***C.*** 100 ***D.*** 93 ***E.*** 51

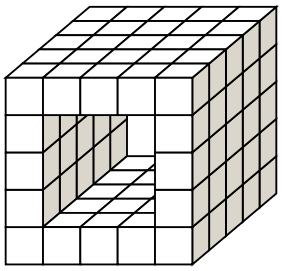
7. Robert built this tunnel out of identical cubes. The hole goes straight through from the front to the back. Robert wants to fill the hole completely, so that the tunnel becomes one solid block.



How many more cubes are needed?

A. 8 ***B.*** 12 ***C.*** 36 ***D.*** 40 ***E.*** 48

8. Tom built this tunnel out of identical cubes. The hole goes straight through from the front to the back. Tom wants to fill the hole completely, so that the tunnel becomes one solid block.

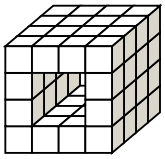


How many more cubes are needed?

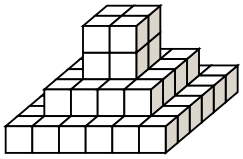
A. 89 ***B.*** 45 ***C.*** 36 ***D.*** 15 ***E.*** 9

Part C: Each question is worth 5 points

9. Leo has the tunnel on the left. Leo takes it apart to build the pyramid on the right, but there are not enough cubes. Nothing is hollow inside the pyramid.



tunnel

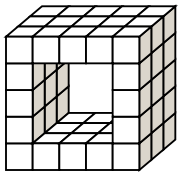


pyramid

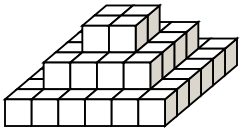
How many more cubes does Leo need?

A. 8 ***B.*** 9 ***C.*** 11 ***D.*** 12 ***E.*** 48

10. Leo has the tunnel on the left. Leo takes it apart to build the pyramid on the right, but there are not enough cubes. Nothing is hollow inside the pyramid.



tunnel

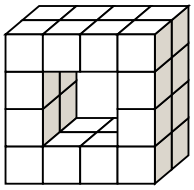


pyramid

How many more cubes does Leo need?

A. 8 ***B.*** 7 ***C.*** 6 ***D.*** 5 ***E.*** 4

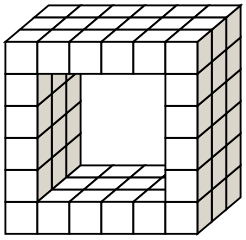
11. Sara's tunnel is 2 cubes long. The hole goes straight through. Sara makes the tunnel longer, keeping the same front, until it is 5 cubes long.



How many cubes does Sara have to add?

A. 12 ***B.*** 24 ***C.*** 36 ***D.*** 48 ***E.*** 60

12. Robert's tunnel is 3 cubes long. The hole goes straight through. Robert makes the tunnel longer, keeping the same front, until it is 5 cubes long.



How many cubes does Robert have to add?

A. 40 ***B.*** 36 ***C.*** 31 ***D.*** 20 ***E.*** 12

Answer Key

1. **D**	2. **C**	3. **B**	4. **E**	5. **E**	6. **D**
7. **B**	8. **B**	9. **D**	10. **A**	11. **C**	12. **A**

Solutions

1. D (36) — The front has 4 × 4 – 2 × 2 = 12 cubes. The tunnel is 3 cubes long, so 12 × 3 = 36. [2002 Q20 · count_tunnel]

2. C (64) — The front has 5 × 5 – 3 × 3 = 16 cubes. The tunnel is 4 cubes long, so 16 × 4 = 64. [2002 Q20 · count_tunnel]

3. B (16) — Tunnel: 12 cubes in the front, 3 long: 12 × 3 = 36. Pyramid: 4 × 4 + 2 × 2 = 20. Left over: 36 – 20 = 16. [2002 Q20 · tunnel_to_pyramid]

4. E (28) — Tunnel: 16 cubes in the front, 4 long: 16 × 4 = 64. Pyramid: 5 × 5 + 3 × 3 + 1 × 1 × 2 = 36. Left over: 64 – 36 = 28. [2002 Q20 · tunnel_to_pyramid]

5. E (44) — Count layer by layer, from the bottom: 5 × 5 + 3 × 3 × 2 + 1 × 1 = 44. [2002 Q20 · count_pyramid]

6. D (93) — Count layer by layer, from the bottom: 7 × 7 + 5 × 5 + 3 × 3 × 2 + 1 × 1 = 93. [2002 Q20 · count_pyramid]

7. B (12) — The hole is 2 × 2 = 4 cubes in the front and goes 3 cubes back: 4 × 3 = 12. [2002 Q20 · fill_the_hole]

8. B (45) — The hole is 3 × 3 = 9 cubes in the front and goes 5 cubes back: 9 × 5 = 45. [2002 Q20 · fill_the_hole]

9. D (12) — Tunnel: 12 × 4 = 48 cubes. Pyramid: 6 × 6 + 4 × 4 + 2 × 2 × 2 = 60. Missing: 60 – 48 = 12. [2002 Q20 · not_enough]

10. A (8) — Tunnel: 16 × 3 = 48 cubes. Pyramid: 6 × 6 + 4 × 4 + 2 × 2 = 56. Missing: 56 – 48 = 8. [2002 Q20 · not_enough]

11. C (36) — Each slice of the tunnel has 4 × 4 – 2 × 2 = 12 cubes. 3 new slices: 12 × 3 = 36. [2002 Q20 · longer_tunnel]

12. A (40) — Each slice of the tunnel has 6 × 6 – 4 × 4 = 20 cubes. 2 new slices: 20 × 2 = 40. [2002 Q20 · longer_tunnel]